# Stakeholder Question Catalogue 06

# Academic Office Occupants

80 office-occupant questions mapped to proposed sensing, current authoritative sources, explicit operation types, evidence quality, uncertainty, permissions and safe answer boundaries.

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| **STAKEHOLDER** | Academic staff and other occupants using Abacws offices, shared academic workrooms and nearby support spaces for desk-based work, meetings, supervision, collaboration, preparation and administration. |
| **QUESTION RANGE** | Focused work and day planning; office comfort, light, air and noise; meetings and supervision; shared desks and resources; hybrid work and dependable services; transitions and visitors; accessibility, well-being, irregular work, disruptions, safety and sustainability. |
| **EVIDENCE MODEL** | Proposed, correctly located sensing is combined with authorised room, allocation, booking, access, accessibility, service, BMS, maintenance, visitor, policy and safety sources. Every record requires current source authority, valid spatial coverage, time alignment, data quality, uncertainty and a safe answer boundary. |

**Core principle:** Support the occupant's own workspace, meeting, route or contingency decision without inspecting work content, identifying or tracking people, inferring presence history or productivity, bypassing responsible authorities, or sounding more certain than the evidence.

# 1. Catalogue scope and design principles

This catalogue covers practical decisions academic office occupants may ask a conversational university building to support during desk work, meetings, supervision, hybrid work, transitions, visitor handling and irregular hours. The service assists the occupant and does not take over room allocation, booking, IT, facilities, accessibility, security or safety responsibilities.

### Proposed capability, not current fact

Every sensor, integration and service is a design requirement unless deployment, room coverage, permissions and operational status are independently verified. The catalogue does not claim that all capabilities currently exist.

### Current building information

The May 2021 floor plan is a historical spatial reference. Current room purpose, ownership, allocation, capacity, staff access, bookability, equipment, routes, lifts and services require authoritative verification.

### Occupant-owned decisions

Questions support choosing when and where to work, preparing meetings, using shared resources, planning routes, handling visitors, adapting to disruptions and making proportionate sustainability choices. They do not make facilities, IT, security or safety decisions.

### Correctly located evidence

A room-level claim requires a sensor inside that room or a validated served zone. Corridor and adjacent-room data may be context but cannot be silently substituted. Timestamps, quality, calibration, gaps and uncertainty remain visible.

### Privacy by design

Use anonymous aggregate occupancy and numerical acoustic indicators only where necessary. Do not inspect emails, documents, meetings, calls, research or teaching content, speech, network traffic or device content; identify people; or infer presence history or productivity.

### Authoritative-source precedence

Room allocation, booking, access, accessibility, services, equipment, maintenance, closures, alarms, lone-working and emergencies come from authorised systems or competent authorities. Sensor data cannot grant permission or certify safety.

### Honest operation and failure

Each record labels observation, lookup, calculation, estimate, forecast, comparison, diagnosis or recommendation. Missing, stale, conflicting or unauthorised evidence produces clarification, a limited answer, deferral or not assessable.

### Stakeholder value

The output should help an office occupant make a practical work, meeting, route or contingency decision, not create an operational dashboard or shift staff responsibilities onto the conversational service.

# 2. How to read each question record

Every record turns a natural office-occupant question into a bounded evidence requirement. The fields identify the proposed sensing, current authoritative source, permitted operation, uncertainty and the point at which the service must qualify, defer or abstain.

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| **OFFICE-OCCUPANT QUESTION** | A natural question owned by an office occupant. It is phrased around a workspace, meeting, supervision, hybrid-work, route, accessibility, disruption or sustainability decision rather than an operator task. |
| **SENSORS AND TELEMETRY** | Physically plausible proposed room, desk-zone, route, network, power, BMS or service evidence. The field states target-room, desk-zone, route or served-asset placement, timing, aggregation and commissioning/calibration conditions relevant to the question. |
| **AUTHORITATIVE SUPPORTING DATA** | Current room purpose, allocation, booking, access, accessibility, equipment, IT, maintenance, visitor, policy, lone-working or safety records that must take precedence over sensor inference. |
| **ANALYSIS REQUIRED** | The exact operation: observation, lookup, calculation, estimate, forecast, comparison, diagnosis or recommendation. It identifies the decision interval, minimum evidence, comparator or baseline, uncertainty and failure behaviour. |
| **ANSWER BOUNDARY** | What the building must not infer, disclose, control or certify; when it must ask for clarification, provide a limited answer, defer to an authority or return not assessable. |
| **WHY THIS IS USEFUL** | The concrete office-occupant decision supported by the answer. |

### Readiness

**R1 - practical**

Core proposed sensing plus basic current spatial, service or provenance data.

**R2 - integration-dependent**

Room, allocation, booking, access, accessibility, IT, BMS, maintenance, visitor, lone-working or forecasting integration.

**R3 - research-grade or restricted**

Dense sensing, validated comparison/diagnosis, controlled operational data, or stronger privacy, governance and safety controls.

**Global rule:** use only verified eligible spaces and authorised data; preserve timestamps and room-level coverage; inspect no email, document, meeting, call, speech or network content; never infer identity, presence history or productivity; and return a limited or not-assessable answer when critical evidence is missing.

# 3. Worked example: a mixed office day

A responsible answer combines user-authorised commitments and office requirements with verified room, access, service, accessibility and environmental evidence. It never treats an empty room as available, a quiet room as private, or environmental telemetry as permission or safety certification.

**Office occupant:** "I need two hours of quiet writing, a confidential supervision meeting, a hybrid call and an evening work period. Can you identify suitable authorised spaces, service risks, accessible routes, verified fallbacks and where the evidence is too weak to recommend?"

### 1. Resolve the day

Use only the occupant's authorised times, locations and requirements for focused work, supervision, hybrid service, access, privacy, accessibility and late working.

### 2. Verify authoritative status

Use current room purpose, allocation, booking, access, accessibility, service, visitor, lone-working, maintenance and closure records before consulting sensors.

### 3. Verify evidence quality

Require target-room or validated served-zone coverage, timestamps, calibration/commissioning, uptime, gaps, drift and current service-health provenance. Mark unverified criteria unknown.

### 4. Perform the right operation

Use lookup for permissions/status, observation for current conditions, calculation for routes, forecast for future risk and diagnosis only when time-aligned evidence supports it.

### 5. Rank occupant-owned actions

Recommend work windows, approved rooms, route buffers, pre-checks, rechecks and verified fallbacks; do not book, grant access, operate systems or take over IT, facilities or safety decisions.

### 6. Communicate answerability

State sources, observed/retrieved times, confidence, omitted criteria and safe failure. Return a conditional or not-assessable result when essential evidence or authority is missing.

**ILLUSTRATIVE ANSWER STRUCTURE**

"The 9-11 a.m. writing option is authorised and currently quiet, but forecast confidence is moderate because occupancy coverage has a gap. The supervision room is confirmed available and access-approved; privacy is based on its authoritative classification, not silence. The hybrid room reports network and power ready at [time], but personal-device compatibility remains unverified. The evening work period is conditional on current access and lone-working requirements. Recheck lift, closure and service status at [time]."

## 1. Focused desk work and day planning

Questions AO-001-AO-002

### AO-001

CORE | L2 - forecast + recommendation | R2 - integration-dependent

**Academic office occupant question:** I have three hours for writing before a meeting. Which workspace I am authorised to use is most likely to stay quiet, comfortable and available throughout?

#### Sensors and telemetry

in-room temperature, RH, direct NDIR CO2, illuminance, numerical acoustic indicators and anonymous occupancy in every eligible workspace. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

#### Authoritative supporting data

Authoritative sources: current workspace allocation, permitted use and bookability; opening and access status; the meeting time/location supplied by the user; accessible routes; closures and service incidents. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authorised lookup, forecast and multi-criteria recommendation. Filter to spaces the user may use for the full interval, predict quietness and environmental stability from comparable periods, then rank by fit and walking time. Decision interval: the stated three-hour work period. Use eligible comparable history and known schedules; report range, confidence and recheck time, apply permission/accessibility/privacy/safety constraints before ranking, and provide only a verified fallback.

#### Answer boundary

An empty or quiet-looking space is not automatically available or private. Do not inspect calendar content beyond the user-authorised time and destination. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant protect a substantial writing block and select a genuinely authorised fallback before the day begins.

### AO-002

CORE | L2 - temporal comparison | R2 - integration-dependent

**Academic office occupant question:** My shared office is usually busy after lunch. What is the best remaining time today for uninterrupted desk work?

#### Sensors and telemetry

anonymous room occupancy, doorway counts and numerical LAeq/percentile sound metrics inside the shared office, plus room temperature and CO2 for context. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

#### Authoritative supporting data

Authoritative sources: current staff access and workspace-allocation rules; the user-authorised working window; building opening, closures and known events. No personal movement history or identifiable access log is required. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: current observation, short-horizon forecast and comparison. Compare candidate time windows using recent same-day trends and matched weekdays, then estimate the likelihood of lower occupancy and noise while keeping environmental suitability visible. Decision interval: the requested period today. Use eligible comparable history, aligned clocks/windows and relevant controls; report prediction range, confidence, effect and recheck time without claiming causality.

#### Answer boundary

Do not identify who is present, infer individual work patterns or promise silence. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant choose a lower-interruption work window without profiling colleagues or relying on anecdote.

# 1. Focused desk work and day planning

Questions AO-003-AO-004

## AO-003

CORE | L2 - constrained recommendation | R2 - integration-dependent

**Academic office occupant question:** I need to do confidential administration. Which approved space is formally suitable, rather than merely empty or quiet?

### Sensors and telemetry

room-level anonymous occupancy and door-state telemetry may confirm broad use state; numerical acoustic measurements may describe background sound but cannot prove speech privacy. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

### Authoritative supporting data

Authoritative sources: room purpose, staff access, booking, capacity, privacy classification and permitted activity; current closures and accessibility information. These records are mandatory before any environmental evidence is considered. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

### Analysis required

Operation: authorised lookup and recommendation. Apply permission and privacy suitability as hard constraints, then compare verified options by availability, access, environmental conditions and route. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

### Answer boundary

Low occupancy, a closed door or low sound level does not certify confidentiality. The service cannot book, grant access or judge information sensitivity. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports confidential administration in a legitimately suitable space without exposing content or encouraging unauthorised room use.

## AO-004

CORE | L2 - schedule synthesis | R2 - integration-dependent

**Academic office occupant question:** I will be in Abacws from 10:00 to 15:00. Can you fit in focused work, one meeting and preparation time with realistic walking and setup buffers?

### Sensors and telemetry

room-level temperature, $CO_2$, numerical acoustics and anonymous occupancy for eligible workspaces, plus authorised lift/door/closure status and anonymous one-minute flow counts on covered route segments. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

### Authoritative supporting data

Authoritative sources: only the schedule times/locations the user shares; current room assignments, workspace permission, staff access, accessible routes, lift/closure status and opening hours. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

### Analysis required

Operation: authorised lookup, route calculation and recommendation. Build a feasible sequence around fixed commitments, insert suitable work blocks, calculate transition buffers and rank alternatives by walking, access and predicted workspace quality. Decision interval: 10:00-15:00 on the stated day. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

### Answer boundary

The service must not reschedule meetings, expose their content or assume a room is usable outside the authoritative access window. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Turns a short campus visit into a feasible sequence with realistic work blocks and transition time.

# 1. Focused desk work and day planning

Questions AO-005-AO-006

## AO-005

USEFUL | L2 - zone comparison | R3 - research-grade or restricted

**Academic office occupant question:** In my allocated shared office, which permitted desk zone is most likely to have stable light and temperature this afternoon?

### Sensors and telemetry

desk-zone illuminance, air temperature and optional globe-temperature sensors at representative occupied positions; solar and outdoor temperature reference; anonymous zone occupancy. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

### Authoritative supporting data

Authoritative sources: current desk or zone allocation and permitted seating rules; verified workspace geometry and façade orientation; blind or shading status if authoritatively available; current closures and accessibility needs supplied by the user. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

### Analysis required

Operation: observation, short-horizon forecast and within-room comparison. Compare only commissioned desk zones, account for solar exposure and occupancy, and rank stability rather than a single instantaneous value. Decision interval: the requested period this afternoon. Use eligible comparable history, aligned clocks/windows and relevant controls; report prediction range, confidence, effect and recheck time without claiming causality.

### Answer boundary

Do not infer exact seat availability, control blinds or recommend an unallocated desk. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant choose among permitted desk zones when local light or thermal gradients matter.

## AO-006

CORE | L2 - threshold forecast | R2 - integration-dependent

**Academic office occupant question:** I am beginning a two-hour writing block. Is this office likely to become persistently noisy or poorly ventilated before I finish?

### Sensors and telemetry

in-room NDIR $CO_2$, temperature, RH, numerical acoustic indicators and anonymous occupancy; HVAC airflow or fan status if mapped to the served zone. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

### Authoritative supporting data

Authoritative sources: user-selected writing interval and operational preferences; current room allocation and access; authoritative BMS mapping; planned events, maintenance and closures. Preferences are not medical thresholds. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

### Analysis required

Operation: forecast and alert recommendation. Predict room-level $CO_2$ and acoustic conditions over the interval from current trajectory and comparable periods, state the selected operational criteria and schedule a recheck when uncertainty is high. Decision interval: the stated two-hour work period. Use eligible comparable history and known schedules; report range, confidence and recheck time, apply permission/accessibility/privacy/safety constraints before ranking, and provide only a verified fallback.

### Answer boundary

Do not record speech, infer who is making noise or claim that $CO_2$ alone measures complete air quality. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports an early decision to continue, adjust the writing block or move to a verified alternative.

## 1. Focused desk work and day planning

Questions AO-007-AO-008

### AO-007

CORE | L2 - service-risk recommendation | R2 - integration-dependent

**Academic office occupant question:** I have a deadline tomorrow. Which approved workspace has the strongest verified combination of power, network and environmental reliability for a long session?

#### Sensors and telemetry

target-room temperature, $CO_2$, illuminance, numerical acoustics and anonymous occupancy; authorised room-circuit or outlet status; wireless access-point health and service incidents. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

#### Authoritative supporting data

Authoritative sources: current workspace permission, opening/access hours, room/service inventory, network and power status, planned work, closures and accessible routes. Show affected scope, source owner, effective and retrieval times and permission. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authorised lookup, reliability comparison and recommendation. Filter eligible spaces, compare recent service availability and current environment, then rank options with separate confidence for power, network and environmental evidence. Decision interval: the stated period tomorrow. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

#### Answer boundary

Do not promise uninterrupted service, inspect network traffic or infer workstation performance. If room/asset mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback; never promise device-specific performance.

**Why this is useful:** Helps the occupant choose a robust workspace for deadline-critical work and understand remaining service risk.

### AO-008

USEFUL | L2 - historical forecast | R2 - integration-dependent

**Academic office occupant question:** On comparable Wednesdays, is this office usually calmer before 09:00 or after 16:00?

#### Sensors and telemetry

anonymous room occupancy, doorway counts and numerical acoustic statistics inside the office or validated served workspace. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

#### Authoritative supporting data

Authoritative sources: authorised opening and staff-access periods; current term and event calendar at aggregate level; closures and changes in room use. No individual access history is needed. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: historical comparison with forecast interpretation. Compare matched Wednesdays across the relevant term, control for holidays and major events, and report median conditions, variability and confidence rather than a single average. Decision interval: the compared early and late Wednesday windows. Use time-aligned mapped sources and eligible controls; retain alternative explanations, report uncertainty, and return no defensible diagnosis if evidence cannot distinguish them.

#### Answer boundary

Do not identify regular users or present historical patterns as a guarantee for the next Wednesday. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports a realistic choice between early and late office work without identifying regular users.

## 1. Focused desk work and day planning

Questions AO-009-AO-010

### AO-009

CORE | L2 - accessibility-aware recommendation | R2 - integration-dependent

**Academic office occupant question:** Where can I alternate focused work with short step-free breaks while keeping an authorised workspace available?

#### Sensors and telemetry

room-level environment, anonymous occupancy and numerical acoustic indicators in eligible workspaces; lift and route status from authorised systems. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current workspace permissions and bookability; authoritative accessibility information, route graph, lift status, opening hours, break-space classification and closures; user-stated access preferences. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authorised lookup, route calculation and multi-criteria recommendation. Apply access and permitted-use constraints first, then rank workspaces by predicted quietness, environmental quality and distance to a verified break area. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

The service cannot certify accessibility from sensors or infer health needs. It must not disclose occupancy of sensitive spaces. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Helps the occupant plan sustainable work-and-break cycles while preserving access and workspace permissions.

### AO-010

USEFUL | L3 - contextual diagnosis | R2 - integration-dependent

**Academic office occupant question:** This office is unusually distracting today. What room-level conditions differ from normal, and which authorised alternative is genuinely usable?

#### Sensors and telemetry

room-level numerical acoustics, anonymous occupancy, temperature, CO2, illuminance and door-state context, with matched nearby workspace evidence only where each room is separately instrumented and commissioned. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

#### Authoritative supporting data

Authoritative sources: current room allocation and authorised alternatives; bookings, access, closures, known works and event schedule; user-reported time and type of distraction without collecting conversation content. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: observation, contextual comparison and recommendation. Compare the current interval with the rooms normal matched baseline, identify which measured factors are unusual, then rank only verified alternatives. Decision interval: the requested period today. Use time-aligned mapped sources and eligible controls; retain alternative explanations, report uncertainty, and return no defensible diagnosis if evidence cannot distinguish them.

#### Answer boundary

A subjective feeling may not be explained by available sensors. Do not analyse speech or infer behaviour. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports an evidence-aware choice to stay, change the task or move to an approved alternative.

## 2. Office environmental comfort and concentration

Questions AO-011-AO-012

### AO-011

CORE | L2 - matched comparison | R3 - research-grade or restricted

**Academic office occupant question:** Is my office significantly warmer than comparable offices today once sunlight, occupancy and outdoor conditions are considered?

#### Sensors and telemetry

calibrated in-room air temperature, optional globe temperature, illuminance/solar proxy and anonymous occupancy in the target and separately instrumented comparable offices, plus outdoor weather. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

#### Authoritative supporting data

Authoritative sources: current room purpose, façade/shading context, served-zone mapping and maintenance state; comparison offices must have equivalent use and valid sensing. Preserve source owner, effective/retrieval time, permission and unresolved differences. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: calculation and matched spatial comparison. Compare the same interval, adjust descriptively or statistically for occupancy and solar exposure, and report effect size, variability, missingness and whether the difference persists. Decision interval: the requested period today. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

#### Answer boundary

Do not expose other occupants or silently use corridor data. A warmer reading does not prove a fault. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant distinguish a real room-specific thermal pattern from weather, sunlight or occupancy effects.

### AO-012

USEFUL | L3 - multi-factor interpretation | R3 - research-grade or restricted

**Academic office occupant question:** The office feels stuffy even though $CO_2$ is not high. Which other measured conditions could plausibly explain the feeling?

#### Sensors and telemetry

in-room direct NDIR $CO_2$, temperature, RH, PM2.5, indicative TVOC, air speed and supply-airflow telemetry mapped to the served zone; sensor health and outdoor reference. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only.

#### Authoritative supporting data

Authoritative sources: current HVAC served-zone and operating-state mapping, room use and known maintenance; user-reported time and sensation. Any health or exposure assessment remains outside the conversational service. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: diagnostic synthesis. Verify $CO_2$ and sensor quality, examine thermal, humidity, particulate, VOC-indicator and airflow patterns, compare with matched baseline and rank plausible measured contributors without claiming causality. Decision interval: the user-specified interval. Use time-aligned mapped sources and eligible controls; retain alternative explanations, report uncertainty, and return no defensible diagnosis if evidence cannot distinguish them.

#### Answer boundary

Normal $CO_2$ does not certify good air quality, and low-cost TVOC is not compound-specific. Do not give medical advice. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant decide whether to use an approved ventilation option, relocate temporarily or report a better-defined issue.

## 2. Office environmental comfort and concentration

Questions AO-013-AO-014

### AO-013

CORE | L2 - recurrence analysis | R2 - integration-dependent

**Academic office occupant question:** Has elevated CO2 recurred during occupied hours in this office over the past month, or is today unusual?

**Sensors and telemetry**

in-room calibrated NDIR CO2 and anonymous occupancy, with one-minute records, sensor-health, relocation, calibration and missing-data metadata. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

**Authoritative supporting data**

Authoritative sources: current room purpose and valid monitoring period; authorised HVAC served-zone mapping and maintenance events; the operational criterion selected for analysis. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: historical calculation and anomaly comparison. Identify sustained episodes using a stated duration rule, compare today's duration and peak with the valid prior month, and report event count, data completeness and uncertainty. Decision interval: the stated one-month historical window. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

**Answer boundary**

Do not treat a single spike as a sustained event or use CO2 as a complete air-quality measure. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** This distinguishes a recurring ventilation-related pattern from a one-off event and supports proportionate escalation.

### AO-014

USEFUL | L2 - solar/glare forecast | R3 - research-grade or restricted

**Academic office occupant question:** Is glare at my permitted desk likely to recur this week, and during which afternoon hours is the risk highest?

**Sensors and telemetry**

desk-plane illuminance and vertical illuminance or glare proxy at the actual occupied position, outdoor solar radiation, blind state if authorised and calibrated, and local geometry/orientation. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only.

**Authoritative supporting data**

Authoritative sources: current desk allocation, façade and shading geometry, authorised blind-operating rules, weather forecast and office access. Historic floor-plan orientation alone is insufficient. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: observation, historical comparison and forecast. Detect sustained high-contrast periods from validated desk-position data, combine with solar forecast and compare analogous days. Decision interval: the stated period this week. Use eligible comparable history, aligned clocks/windows and relevant controls; report prediction range, confidence, effect and recheck time without claiming causality.

**Answer boundary**

A ceiling lux sensor cannot establish desk-level glare. Do not use a camera to analyse the occupant. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant choose a better desk position or work period before recurring glare disrupts desk work.

## 2. Office environmental comfort and concentration

Questions AO-015-AO-016

### AO-015

CORE | L2 - source-context comparison | R2 - integration-dependent

**Academic office occupant question:** Is today's noise a brief corridor-related peak or a sustained room-level problem likely to continue?

#### Sensors and telemetry

numerical LAeq and percentile sound metrics inside the office and at a separate corridor reference point; door-state and anonymous corridor-flow counts. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current room and corridor mapping, known works or events, door status and user-reported interval. No individual identity or conversation data. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: observation and time-aligned comparison. Separate transient peaks from sustained room background, compare office and corridor timing, and report whether the pattern is consistent with corridor activity, room sources or unresolved evidence. Decision interval: the requested period today. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

#### Answer boundary

Acoustic correlation does not identify a person or prove a source. Do not record speech. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports a practical choice between waiting out a short peak, changing the task or moving to a verified alternative.

### AO-016

CORE | L2 - multi-objective forecast | R2 - integration-dependent

**Academic office occupant question:** Which remaining period today has the best supported balance of temperature, ventilation, light and quiet for concentrated work?

#### Sensors and telemetry

in-room temperature, RH, NDIR $CO_2$, $PM_{2.5}$, illuminance, numerical acoustic statistics and anonymous occupancy. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

#### Authoritative supporting data

Authoritative sources: current office access, room allocation, opening hours, planned events, maintenance and local weather forecast; occupant-selected acceptable ranges and working window. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: short-horizon forecast and multi-criteria recommendation. Predict each variable for candidate intervals, apply hard access constraints, score trade-offs transparently and show confidence by criterion. Decision interval: the requested period today. Use eligible comparable history and known schedules; report range, confidence and recheck time, apply permission/accessibility/privacy/safety constraints before ranking, and provide only a verified fallback.

#### Answer boundary

The result is a forecast, not a guarantee or health judgement. Do not infer personal productivity. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant schedule demanding work during the strongest combined environmental and acoustic window.

## 2. Office environmental comfort and concentration

Questions AO-017-AO-018

### AO-017

USEFUL | L3 - local airflow diagnosis | R3 - research-grade or restricted

**Academic office occupant question:** I feel a cold draught at my desk. Do local airflow and nearby door or window readings support that explanation?

#### Sensors and telemetry

low-speed air-velocity sensing at the occupied desk zone, air and globe temperature, door/window contacts and supply-airflow or diffuser telemetry mapped to the served zone. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only.

#### Authoritative supporting data

Authoritative sources: current desk position and room geometry, verified diffuser/door/window locations, HVAC mapping, authorised window state and maintenance history. The May 2021 plan alone cannot establish current airflow paths. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: observation and cautious diagnosis. Compare draught time with local air speed, temperature, door/window events and supply-airflow changes, then rank supported explanations and unresolved alternatives. Decision interval: the user-specified interval. Use time-aligned mapped sources and eligible controls; retain alternative explanations, report uncertainty, and return no defensible diagnosis if evidence cannot distinguish them.

#### Answer boundary

Do not infer a mechanical fault from association or recommend unauthorised equipment changes. If room/asset mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback; never promise device-specific performance.

**Why this is useful:** Provides evidence for changing permitted desk position or reporting a repeatable local draught problem.

### AO-018

CORE | L2 - indoor/outdoor comparison | R3 - research-grade or restricted

**Academic office occupant question:** Are particulate levels in this office persistently higher than outdoors and in separately monitored comparable offices?

#### Sensors and telemetry

calibrated or correction-validated PM2.5 and PM10 sensors inside the target office, at an outdoor reference and in separately instrumented comparable offices; ventilation state and aligned one-minute records. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only.

#### Authoritative supporting data

Authoritative sources: current room purpose, valid outdoor-reference location, HVAC served-zone mapping, known works and cleaning events. Comparison spaces must be authorised for aggregate analysis and environmentally comparable. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: calculation and time-aligned spatial comparison. Estimate sustained indoor/outdoor ratios, event duration and uncertainty; examine ventilation and local-event context without assigning a source unless evidence supports it. Decision interval: the user-specified interval. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

#### Answer boundary

Low-cost particle sensors are sensitive to calibration and humidity. Do not make medical or compliance claims. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant judge whether particulate conditions justify temporary relocation or a documented investigation.

## 2. Office environmental comfort and concentration

Questions AO-019-AO-020

### AO-019

USEFUL | L2 - historical range analysis | R2 - integration-dependent

**Academic office occupant question:** During my normal work hours, has this office repeatedly been unusually dry or humid this month?

**Sensors and telemetry**

calibrated in-room RH and temperature with one-minute records, health flags and sufficient coverage; outdoor humidity for context. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only.

**Authoritative supporting data**

Authoritative sources: user-defined working-hour window, current room allocation and monitoring-validity period; HVAC maintenance and known unusual events. No personal attendance history is needed. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: historical calculation and comparison. Summarise median, range, duration outside the selected operational band and comparison with matched outdoor conditions, while reporting missingness and sensor uncertainty. Decision interval: the stated one-month historical window. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

**Answer boundary**

Humidity alone does not diagnose symptoms or establish health risk. Do not infer when the user was present. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports work planning and a proportionate service request when dryness or humidity is persistent rather than occasional.

### AO-020

USEFUL | L2 - preference comparison | R3 - research-grade or restricted

**Academic office occupant question:** Can you compare the room conditions with comfort feedback I chose to provide for the same periods, without creating a health or productivity profile?

**Sensors and telemetry**

in-room temperature, RH, CO2, air speed, radiant temperature, light and numerical acoustics as relevant; voluntary timestamped feedback linked only to the requested session and retained under an approved policy. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: numerical sound only, with no speech or raw audio.

**Authoritative supporting data**

Authoritative sources: user consent and feedback-retention rules; current room and sensor mapping; no emails, documents, work output, attendance or health records. Access to feedback is restricted to the user and approved service purpose. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: observation and consent-bounded descriptive comparison. Align submitted sensations with measured conditions, show which variables were observed and whether similar patterns recur, without predictive profiling unless separately consented and validated. Decision interval: the user-specified interval. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

**Answer boundary**

Do not infer medical status, productivity, protected characteristics or general preference from sparse feedback. Allow deletion and abstain if consent, linkage, coverage or sample size is inadequate. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant interpret voluntarily submitted comfort feedback without creating a personal evaluation profile.

## 3. Meetings, supervision and collaboration

Questions AO-021-AO-022

### AO-021

CORE | L2 - privacy-aware recommendation | R2 - integration-dependent

**Academic office occupant question:** I need a confidential supervision meeting. Which room I may use is available, step-free and officially classified for the required privacy?

#### Sensors and telemetry

room-level anonymous occupancy and numerical acoustic background indicators may support suitability; door state can show current status but not confidentiality. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: booking, room-purpose, privacy classification, capacity, staff access and accessibility data; current closures and route status; meeting duration and access needs supplied by the user. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authorised lookup and recommendation. Apply privacy classification, access, accessibility and availability as hard constraints, then compare verified environmental background and route. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

Low sound, a closed door or apparent vacancy does not certify privacy. The service cannot book the room or disclose another meeting. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Supports confidential supervision while respecting room authority, access, privacy and accessibility requirements.

### AO-022

CORE | L2 - ventilation forecast | R2 - integration-dependent

**Academic office occupant question:** For a six-person, 90-minute meeting, is the assigned room likely to remain adequately ventilated, and when should I recheck?

#### Sensors and telemetry

calibrated in-room NDIR $CO_2$, temperature, RH and anonymous occupancy, plus supply-airflow or fan status mapped to the room. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

#### Authoritative supporting data

Authoritative sources: current room capacity/purpose, verified booking, user-supplied group size and duration, HVAC served-zone mapping, operating schedule, faults and closures. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: forecast. Use current conditions and comparable meetings or a validated $CO_2$ accumulation model to estimate the rooms trajectory, state assumptions and identify a recheck time or break recommendation. Decision interval: the planned 90-minute meeting and a stated pre-check window. Use eligible comparable history and known schedules; report prediction range, confidence and recheck time, or fall back to confirmed status.

#### Answer boundary

The service cannot guarantee ventilation or certify safety, and $CO_2$ is not complete air quality. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant decide whether to retain the room, add a break or request an approved alternative.

# 3. Meetings, supervision and collaboration

Questions AO-023-AO-024

## AO-023

CORE | L2 - route + setup recommendation | R2 - integration-dependent

**Academic office occupant question:** I have back-to-back supervision meetings. Which verified room option minimises walking and setup time without revealing other people's booking details?

### Sensors and telemetry

lift and broad route-flow status; room-level environment only for eligible options. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

### Authoritative supporting data

Authoritative sources: user-authorised meeting times and destinations; current room availability, staff access, permitted setup windows, capacity, privacy classification, accessibility and closures. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

### Analysis required

Operation: authorised lookup, route calculation and recommendation. Filter to valid rooms, calculate travel and setup buffers, then rank by proximity and verified suitability. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

### Answer boundary

Do not reveal who booked a room, change bookings or assume an apparently empty room is available. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Reduces transition pressure between supervision meetings without exposing another person's booking details.

## AO-024

CORE | L2 - hybrid-room recommendation | R2 - integration-dependent

**Academic office occupant question:** Which eligible room best supports an accessible hybrid meeting with verified network, power and assistive-technology readiness?

### Sensors and telemetry

Authorised room-level network service health without traffic inspection; room power or AV status; proposed numerical acoustic and environmental sensing. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: numerical sound only, with no speech or raw audio; service telemetry mapped to the room/access point/circuit/asset; preserve scope, event time, maintenance state and uptime; inspect no traffic or device content.

### Authoritative supporting data

Authoritative sources: current room features, bookability, staff access, capacity, accessibility and assistive-technology inventory; AV/IT incidents, maintenance and closures; user-supplied meeting requirements. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

### Analysis required

Operation: authoritative lookup, service comparison and recommendation. Treat access, capacity and required assistive technology as hard constraints, then compare current service status, environmental conditions and uncertainty. Decision interval: the user-specified interval. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

### Answer boundary

Do not inspect meeting content, grant access or certify that personal devices will work. If room/asset mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback; never promise device-specific performance.

**Why this is useful:** Helps the occupant select an inclusive hybrid-meeting option while leaving service ownership with AV/IT and accessibility teams.

## 3. Meetings, supervision and collaboration

Questions AO-025-AO-026

### AO-025

USEFUL | L2 - recurrence forecast | R2 - integration-dependent

**Academic office occupant question:** This meeting room became stuffy yesterday. How likely is the same pattern tomorrow for the planned duration and group size?

**Sensors and telemetry**

room-level NDIR CO2, temperature, RH, anonymous occupancy and mapped supply-airflow; one-minute history with calibration, uptime and gap metadata. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

**Authoritative supporting data**

Authoritative sources: current booking, expected group size and duration, HVAC schedule, served-zone mapping, maintenance state, weather forecast and comparable prior meetings in the same room. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: historical comparison and forecast. Reconstruct yesterday's event, identify comparable cases and forecast tomorrow's trajectory with explicit assumptions and confidence. Decision interval: the stated period tomorrow. Use eligible comparable history, aligned clocks/windows and relevant controls; report prediction range, confidence, effect and recheck time without claiming causality.

**Answer boundary**

Do not assume the same attendance or HVAC state. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** This allows the occupant to address a likely recurring room problem before colleagues arrive.

### AO-026

CORE | L2 - activity-space recommendation | R2 - integration-dependent

**Academic office occupant question:** Which authorised space suits a short collaborative discussion without unnecessarily disturbing nearby focused work?

**Sensors and telemetry**

numerical acoustic background indicators and anonymous occupancy in eligible collaboration spaces and nearby work zones; door state and broad route-flow context. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

**Authoritative supporting data**

Authoritative sources: current room purpose, permitted activity, access, availability, capacity and adjacency; quiet-zone policy, opening hours, accessibility and closures. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: authorised lookup, spatial reasoning and recommendation. Exclude quiet or unauthorised areas, then rank verified collaboration spaces by capacity, current use, acoustic separation and walking distance. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

**Answer boundary**

Do not infer that a low-noise area permits discussion or that an empty space is available. The service cannot enforce behaviour. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports brief collaboration while reducing avoidable disturbance to nearby focused-work areas.

## 3. Meetings, supervision and collaboration

Questions AO-027-AO-028

### AO-027

CORE | L2 - visitor meeting-point recommendation | R2 - integration-dependent

**Academic office occupant question:** A visitor is joining me. Which approved meeting point is easy to find, accessible and likely to remain comfortable at the planned time?

#### Sensors and telemetry

public-area temperature, $CO_2$, light, numerical acoustic indicators and aggregate footfall; authorised lift and entrance status. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current visitor process, approved meeting locations, opening and access rules, accessible routes, lift status, temporary closures and appointment destination. The visitor identity is not needed for ranking. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authorised lookup, route calculation and recommendation. Filter to approved public meeting points, compare route simplicity, accessibility and current environment, and provide a clearly described fallback. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

Do not direct visitors into staff-only areas, grant access or disclose sensitive occupancy. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** This improves the visitor's arrival and waiting experience without weakening access control.

### AO-028

USEFUL | L2 - matched-room comparison | R3 - research-grade or restricted

**Academic office occupant question:** Our meeting was repeatedly interrupted by noise. Which authorised room is a better-supported option at the same time next week?

#### Sensors and telemetry

numerical LAeq/percentile sound metrics in candidate rooms and nearby circulation zones, anonymous occupancy and door state; no raw audio. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

#### Authoritative supporting data

Authoritative sources: current room availability, purpose, access, capacity, privacy and accessibility; next-weeks events, known works and route constraints. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: historical comparison, forecast and recommendation. Compare the same weekday/time across eligible rooms, control for known events and occupancy, then rank likely acoustic suitability with uncertainty. Decision interval: the stated time next week. Use eligible comparable history and known schedules; report range, confidence and recheck time, apply permission/accessibility/privacy/safety constraints before ranking, and provide only a verified fallback.

#### Answer boundary

Do not infer the source or content of noise, or promise a disturbance-free room. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports a better meeting-room choice without monitoring speech or assigning responsibility for noise.

## 3. Meetings, supervision and collaboration

Questions AO-029-AO-030

### AO-029

CORE | L2 - suitability lookup | R2 - integration-dependent

**Academic office occupant question:** Is this space formally suitable for office hours without blocking circulation or overstating privacy?

#### Sensors and telemetry

aggregate occupancy and broad route-flow indicators may describe congestion; numerical acoustics may describe background only. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

#### Authoritative supporting data

Authoritative sources: space purpose, permitted activity, capacity, circulation and egress requirements, accessibility, privacy classification, booking/access and current closures. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authoritative lookup plus constrained suitability calculation. Check permission, circulation, accessibility and privacy requirements first, then use current aggregate evidence as context. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

Sensors cannot authorise use or certify egress, privacy or safety. The service must not disclose another users booking. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant choose a legitimate office-hours setting that preserves circulation, access and realistic privacy.

### AO-030

USEFUL | L2 - post-meeting evidence summary | R2 - integration-dependent

**Academic office occupant question:** After the meeting, can you summarise the measured room conditions against our selected operational ranges without interpreting participant behaviour?

#### Sensors and telemetry

room-level temperature, RH, NDIR $CO_2$, PM2.5, light and numerical acoustic metrics; anonymous aggregate occupancy only if authorised. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

#### Authoritative supporting data

Authoritative sources: verified meeting interval and room, user-selected operational ranges, valid monitoring period and sensor provenance. No participant names, attendance or meeting content. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: calculation and evidence summary. Compute duration and magnitude outside selected ranges, display data completeness and source timestamps, and distinguish observed metrics from interpretation. Decision interval: the user-specified interval. Expose inputs, units, assumptions and a range where uncertainty affects the decision.

#### Answer boundary

Do not infer engagement, attendance, health impact or who affected conditions. Operational ranges are not automatically legal or safety thresholds. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Provides a bounded record for improving future meeting choices or reporting a room issue.

## 4. Shared offices, desks and local resources

Questions AO-031-AO-032

### AO-031

CORE | L1 - availability verification | R2 - integration-dependent

**Academic office occupant question:** Is this shared desk officially available to me now, or does it only appear unoccupied?

**Sensors and telemetry**

desk-presence or zone-occupancy sensing may provide anonymous current-use evidence; room doorway counts can support reconciliation. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

**Authoritative supporting data**

Authoritative sources: desk allocation, reservation/check-in, staff eligibility, opening, access and out-of-service records. These sources take precedence over occupancy inference. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: authorised lookup plus current observation. Report a desk as available only when the authoritative allocation system permits use and sensing does not indicate conflicting occupancy; otherwise show reserved, occupied, unknown or unavailable. Decision interval: the current state at retrieval time. Report the observation window, aggregation, units, freshness, valid coverage and excluded channels.

**Answer boundary**

Do not identify the current user, disclose reservation details or grant use. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** This prevents conflict over shared desks by separating official availability from apparent vacancy.

### AO-032

CORE | L2 - desk recommendation | R2 - integration-dependent

**Academic office occupant question:** Which approved shared workspace currently has a verified usable desk, suitable task lighting and confirmed power?

**Sensors and telemetry**

anonymous desk/zone occupancy, desk-plane illuminance and authorised outlet/circuit service status; room temperature and $CO_2$ for context. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

**Authoritative supporting data**

Authoritative sources: current desk reservation and eligibility, workspace access, power inventory, opening hours, accessibility and out-of-service notices. User power needs are supplied explicitly. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: authorised lookup, current observation and recommendation. Apply desk availability, access and power as hard constraints, then rank by light, environment and route. Decision interval: the current state at retrieval time. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

**Answer boundary**

The service cannot guarantee a particular socket or personal charger will work, and apparent vacancy is insufficient. If room/asset mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback; never promise device-specific performance.

**Why this is useful:** Helps the occupant find a legitimate, functional shared workpoint quickly.

## 4. Shared offices, desks and local resources

Questions AO-033-AO-034

### AO-033

USEFUL | L2 - occupancy-pattern comparison | R2 - integration-dependent

**Academic office occupant question:** At the times I normally use it, has this shared office repeatedly operated near or above its authorised capacity?

#### Sensors and telemetry

anonymous room occupancy and doorway counts, commissioned against room capacity, with one-minute aggregates, uptime and missingness. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

#### Authoritative supporting data

Authoritative sources: current room capacity and purpose, user-specified time windows, valid monitoring period, term calendar and known events. Current capacity must come from an authorised source. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: historical calculation and comparison. Compare occupancy distribution with authorised capacity across the supplied windows, report frequency and duration of high-load periods, and separate data gaps from low use. Decision interval: the user-specified interval. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

#### Answer boundary

Do not infer that the user or specific colleagues were present, and do not treat a sensor estimate as an official headcount. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports a decision to change work hours or request a different authorised shared-work arrangement.

### AO-034

USEFUL | L2 - within-room recommendation | R3 - research-grade or restricted

**Academic office occupant question:** Within this shared office, which permitted desk zone is usually quieter without tracking who uses it?

#### Sensors and telemetry

zone-level numerical acoustic metrics and anonymous zone occupancy, with desk/zone geometry and no raw audio. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

#### Authoritative supporting data

Authoritative sources: current desk allocation, permitted hot-desk zones, accessibility requirements and room layout from an authoritative source. No named seating history. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: current observation, historical comparison and recommendation. Compare permitted zones by recent and matched-period acoustic conditions and occupancy, then rank options while showing uncertainty. Decision interval: the user-specified interval. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

#### Answer boundary

Do not identify usual occupants, infer conversations or recommend an allocated desk. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant choose among permitted desk zones for concentration without tracking colleagues.

## 4. Shared offices, desks and local resources

Questions AO-035-AO-036

### AO-035

USEFUL L2 - queue forecast R2 - integration-dependent

**Academic office occupant question:** What time this afternoon is least likely to be congested around the printer and other shared resources?

#### Sensors and telemetry

anonymous zone occupancy or doorway counts near shared-resource areas; authorised device-service status and queue state where the system exposes it. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

#### Authoritative supporting data

Authoritative sources: current resource location, staff access, operating status, planned maintenance and opening; user-supplied target window. Device management remains authoritative for readiness. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authorised lookup and short-horizon crowding forecast. Combine current aggregate use with comparable time windows, report likely busy periods and identify an alternative authorised resource. Decision interval: the requested period this afternoon. Use eligible comparable history and known schedules; report prediction range, confidence and recheck time, or fall back to confirmed status.

#### Answer boundary

Do not reveal who is printing, inspect print jobs or guarantee immediate availability. If room/asset mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback; never promise device-specific performance.

**Why this is useful:** Reduces avoidable waiting and helps the occupant time routine printing or shared-resource use.

### AO-036

USEFUL L2 - thermal-zone comparison R3 - research-grade or restricted

**Academic office occupant question:** Are temperature differences across this shared office persistent and large enough to justify moving to another permitted desk zone?

#### Sensors and telemetry

multiple calibrated air-temperature and optional globe-temperature sensors across occupied desk zones, plus supply-airflow, solar/light context and anonymous zone occupancy. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

#### Authoritative supporting data

Authoritative sources: current permitted desk zones, room geometry, HVAC served-zone mapping, façade orientation and maintenance status. No inference from the historical plan alone. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: spatial calculation and recommendation. Quantify persistent zone differences over the requested interval, account for solar and occupancy context, and recommend only permitted alternatives with confidence. Decision interval: the user-specified interval. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

#### Answer boundary

Do not create zone conclusions from one sensor or recommend moving another persons belongings. A temperature difference does not prove a fault. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports a practical desk-zone choice when a large shared office has measurable thermal differences.

## 4. Shared offices, desks and local resources

Questions AO-037-AO-038

### AO-037

CORE | L2 - confidential-call suitability | R2 - integration-dependent

**Academic office occupant question:** Which nearby authorised space is suitable for a brief confidential call without recording or analysing speech?

**Sensors and telemetry**

anonymous occupancy and numerical background-sound indicators may describe broad suitability; no microphones retaining audio, speech recognition or conversation analysis. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

**Authoritative supporting data**

Authoritative sources: space purpose, privacy classification, staff access, availability, capacity and accessibility; current closures and permitted use. The user supplies the approximate duration. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: authorised lookup and recommendation. Apply privacy classification and permission first, then compare verified availability, background sound and route. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

**Answer boundary**

Low sound does not certify confidentiality. The service cannot reserve a room or inspect the call. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant handle a short sensitive call without recording content or disturbing others.

### AO-038

USEFUL | L1 - booth status | R2 - integration-dependent

**Academic office occupant question:** Is this shared meeting booth officially available and service-ready now, rather than simply quiet?

**Sensors and telemetry**

anonymous booth occupancy or door/presence state, room environment and authorised power/network status. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; service telemetry mapped to the room/access point/circuit/asset; preserve scope, event time, maintenance state and uptime; inspect no traffic or device content.

**Authoritative supporting data**

Authoritative sources: reservation, permitted-use, access, capacity, equipment status and out-of-service records. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: authorised lookup plus current observation. Return available only when reservation policy and occupancy evidence agree; otherwise report reserved, occupied, degraded, unknown or unavailable, with timestamps. Decision interval: the current state at retrieval time. Report the observation window, aggregation, units, freshness, valid coverage and excluded channels.

**Answer boundary**

Do not reveal the current user or meeting details, and do not infer vacancy from silence. Missing or conflicting authoritative status requires an unknown result or manual check. If room/asset mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback; never promise device-specific performance.

**Why this is useful:** Provides a trustworthy quick check for a shared booth without treating silence as vacancy or privacy.

## 4. Shared offices, desks and local resources

Questions AO-039-AO-040

### AO-039

USEFUL | L2 - longitudinal pattern analysis | R2 - integration-dependent

**Academic office occupant question:** Has aggregate use of this shared office changed since the start of term, and how reliable is the evidence?

**Sensors and telemetry**

anonymous room occupancy and doorway counts with stable commissioning, sensor-change logs, one-minute aggregation and missing-data metadata. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

**Authoritative supporting data**

Authoritative sources: current term dates, room purpose and capacity, known policy or allocation changes, closures and major events. No personal access or identity data. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: historical comparison and change-point analysis. Compare valid pre/post periods, normalise by open hours and known closures, quantify change and report uncertainty, sensor changes and alternative explanations. Decision interval: the current term-to-date window. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

**Answer boundary**

Do not attribute the change to named groups or claim causality from a temporal shift alone. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** This helps occupants understand whether crowding reflects a sustained change rather than a few memorable days.

### AO-040

CORE | L2 - fallback recommendation | R2 - integration-dependent

**Academic office occupant question:** If my allocated desk area is temporarily unavailable, which nearby authorised workspace is the best verified fallback for this task?

**Sensors and telemetry**

room-level environment, anonymous occupancy, authorised network/power status and lift/route state for eligible alternatives. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; service telemetry mapped to the room/access point/circuit/asset; preserve scope, event time, maintenance state and uptime; inspect no traffic or device content.

**Authoritative supporting data**

Authoritative sources: current desk-area outage or closure, staff workspace eligibility, bookability, access, opening, accessibility and service inventory; authoritative incident status. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: authoritative lookup, route calculation and recommendation. Verify the outage, filter eligible alternatives, then rank by functional requirements, environment and distance. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

**Answer boundary**

Do not reallocate desks, bypass access rules or treat a low-occupancy area as authorised. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports continuity of desk-based work during a local workspace disruption.

## 5. Hybrid work, calls and dependable digital services

Questions AO-041-AO-042

### AO-041

CORE | L2 - near-term readiness forecast | R2 - integration-dependent

**Academic office occupant question:** I have a video call in 30 minutes. Is this office likely to provide stable network, power and low background noise, and what should I still test myself?

**Sensors and telemetry**

Authorised room-level wireless health and incident telemetry, mapped circuit/power status, numerical sound metrics and anonymous occupancy. Require current room/access-point/circuit mapping, timestamps, service scope, maintenance state, uptime and gap flags; retain no traffic, device content, speech or raw audio.

**Authoritative supporting data**

Authoritative sources: current office assignment and access, network/power service maps, planned incidents, out-of-service notices and approved fallback spaces; the user supplies call duration and required features. Owner-controlled records valid for the interval take precedence; show source owner, effective/retrieval times and permission scope.

**Analysis required**

Operation: authorised lookup, current observation and short-horizon forecast. Assess current service health and recent stability, forecast broad acoustic conditions for the planned call, and return a readiness classification, fallback and recheck time. Report prediction range and confidence; otherwise return confirmed status only.

**Answer boundary**

Do not inspect call content, device traffic or participants, and do not promise device-specific performance. If mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback.

**Why this is useful:** Helps the occupant avoid a predictable call failure and choose a verified fallback before the meeting starts.

### AO-042

CORE | L2 - confidential hybrid fallback | R2 - integration-dependent

**Academic office occupant question:** If the office network remains degraded, which authorised room is the safest practical fallback for a confidential hybrid meeting?

**Sensors and telemetry**

Authorised network, power and AV status, plus numerical background-sound and anonymous occupancy evidence inside each candidate room. Require current room/asset mapping, timestamps, service scope, maintenance state, uptime and gap flags; retain no traffic, device content, speech or raw audio.

**Authoritative supporting data**

Authoritative sources: current network incident, room privacy classification, availability, access, capacity, accessibility, AV features and closures. Use only rooms the requester may use; owner-controlled room and service records take precedence and must show effective/retrieval times and permission scope.

**Analysis required**

Operation: authoritative lookup, service comparison and recommendation. Apply privacy, permission, availability and accessibility first, then compare verified network, power, AV, environmental and route evidence for the requested interval. Align clocks, report unknowns and uncertainty, and provide one verified fallback.

**Answer boundary**

Do not book or grant access, inspect meeting content or infer privacy from silence. If mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback.

**Why this is useful:** Provides a responsible contingency for confidential hybrid work when the normal office connection is unreliable.

## 5. Hybrid work, calls and dependable digital services

Questions AO-043-AO-044

### AO-043

USEFUL | L3 - incident-domain diagnosis | R3 - research-grade or restricted

**Academic office occupant question:** Yesterday's call quality was poor. Do authorised records support a room-level network, power or environmental problem without inspecting the call?

**Sensors and telemetry**

Authorised access-point health, packet-loss or service-quality aggregates at infrastructure level, room-circuit events and power status; optional room environmental temperature for equipment context. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: service telemetry mapped to the room/access point/circuit/asset; preserve scope, event time, maintenance state and uptime; inspect no traffic or device content.

**Authoritative supporting data**

Authoritative sources: user-reported call interval and room; current network/power asset mapping; authorised incident and maintenance records; service telemetry provenance and clock alignment. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: retrospective diagnosis. Align the reported interval with authorised service and power events, compare nearby service areas and rank supported fault domains while showing gaps and alternative explanations. Decision interval: the stated incident window yesterday plus matched baseline periods. Use time-aligned mapped sources and eligible controls; retain alternative explanations, report uncertainty, and return no defensible diagnosis if evidence cannot distinguish them.

**Answer boundary**

The service cannot diagnose the users device, application or remote participant without authorised evidence, and cannot inspect traffic content. If room/asset mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback; never promise device-specific performance.

**Why this is useful:** Helps the occupant decide whether to test a device, move rooms or report a local or wider service problem.

### AO-044

CORE | L1 - disruption notification | R2 - integration-dependent

**Academic office occupant question:** Can you warn me early enough about an authorised planned network, power or building-service interruption so I can change my workday?

**Sensors and telemetry**

No new environmental sensor is required. Privacy/service guard: service telemetry mapped to the room/access point/circuit/asset; preserve scope, event time, maintenance state and uptime; inspect no traffic or device content.

**Authoritative supporting data**

Authoritative sources: user-selected buildings, rooms, time window and notification preferences; official planned-outage, maintenance, closure and access notices; current workspace allocation and permission. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: authorised lookup and rule-based notification. Match official events to the users selected work locations and times, deduplicate notices and show source, affected service, expected window and official contact. Decision interval: the user-specified interval. State source owner, effective/version time, retrieval time and conflicts; do not extrapolate beyond the record.

**Answer boundary**

Do not infer outages from weak sensor anomalies, expose unrelated calendar data or guarantee restoration time. Unofficial or conflicting information must be labelled and deferred to the owning service. If room/asset mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback; never promise device-specific performance.

**Why this is useful:** This lets the occupant adjust travel and campus work before a confirmed service interruption wastes time.

## 5. Hybrid work, calls and dependable digital services

Questions AO-045-AO-046

### AO-045

USEFUL | L2 - acoustic forecast | R2 - integration-dependent

**Academic office occupant question:** Which time today has the strongest evidence for a quieter important call from an approved workspace?

#### Sensors and telemetry

numerical acoustic metrics in the target office and corridor reference zone, aggregate corridor footfall and door state; no raw audio. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current office access, user-supplied call window, known events, teaching transitions and planned works at aggregate level. No named timetables or individual movement records. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: short-horizon forecast and recommendation. Model corridor noise around known transition periods and current trend, compare candidate windows and state uncertainty and likely transient peaks. Decision interval: the requested period today. Use eligible comparable history and known schedules; report range, confidence and recheck time, apply permission/accessibility/privacy/safety constraints before ranking, and provide only a verified fallback.

#### Answer boundary

Do not promise silence or infer who creates noise. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant schedule an important call during a lower-interruption period.

### AO-046

USEFUL | L1 - visual-condition observation | R2 - integration-dependent

**Academic office occupant question:** Is the lighting in this room suitable for a video call based on illuminance and glare risk, without using a camera to analyse me?

#### Sensors and telemetry

desk/face-plane illuminance and optional vertical illuminance at the intended call position; light-state and daylight context. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only.

#### Authoritative supporting data

Authoritative sources: current room access and call position supplied by the user; lighting service status, blind/shading rules and any approved video-call guidance. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: observation and simple calculation. Compare measured illuminance and stability with a declared operational range, identify strong variability or glare risk and suggest a non-control setup check. Decision interval: the user-specified interval. Expose inputs, units, assumptions and a range where uncertainty affects the decision.

#### Answer boundary

Illuminance cannot guarantee image quality or assess appearance. The service must not activate cameras, control lighting without authority or analyse the user. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports a quick, privacy-preserving lighting decision for a video meeting.

## 5. Hybrid work, calls and dependable digital services

Questions AO-047-AO-048

### AO-047

CORE | L1 - equipment/service lookup | R2 - integration-dependent

**Academic office occupant question:** Are the approved display, docking and assistive-technology services in this room currently listed as ready?

**Sensors and telemetry**

Authorised equipment inventory and service/status telemetry where exposed; power and network status. Privacy/service guard: service telemetry mapped to the room/access point/circuit/asset; preserve scope, event time, maintenance state and uptime; inspect no traffic or device content.

**Authoritative supporting data**

Authoritative sources: current room purpose, access, bookability, approved equipment and assistive-technology inventory, service incidents, maintenance and support ownership. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: authoritative lookup. Report each requested feature as present and ready, present but status unknown, unavailable or not applicable, with source and last update. Decision interval: the current state at retrieval time. State source owner, effective/version time, retrieval time and conflicts; do not extrapolate beyond the record.

**Answer boundary**

Do not inspect connected devices, grant access or treat inventory as proof that every item works. Missing current service status must be shown as unknown, not ready. If room/asset mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback; never promise device-specific performance.

**Why this is useful:** This avoids unnecessary room visits and supports an inclusive setup or early request for assistance.

### AO-048

USEFUL | L2 - hybrid-day recommendation | R2 - integration-dependent

**Academic office occupant question:** I am splitting work between home and campus. Which tasks that genuinely require Abacws facilities should I place within verified service hours?

**Sensors and telemetry**

Authorised network, power, equipment and building-service availability; proposed room environment and occupancy for eligible campus workspaces. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; service telemetry mapped to the room/access point/circuit/asset; preserve scope, event time, maintenance state and uptime; inspect no traffic or device content.

**Authoritative supporting data**

Authoritative sources: minimum user-authorised task categories and time constraints; current opening, access, workspace availability, equipment/service inventory, planned outages and travel constraints. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: authorised lookup and recommendation. Match tasks only to declared requirements, identify campus-dependent work and rank feasible time blocks by verified service availability and workspace quality. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

**Answer boundary**

Do not inspect emails, files or calendars beyond explicitly shared constraints, and do not infer productivity. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant reserve campus time for tasks that genuinely depend on verified on-site services.

## 5. Hybrid work, calls and dependable digital services

Questions AO-049-AO-050

### AO-049

USEFUL | L3 - service-scope diagnosis | R3 - research-grade or restricted

**Academic office occupant question:** Does the authorised evidence indicate a room-local connectivity problem, a wider service incident, or no defensible diagnosis?

**Sensors and telemetry**

Authorised infrastructure health at room access point, floor and building service scopes; room power status and timestamped incident events. Privacy/service guard: service telemetry mapped to the room/access point/circuit/asset; preserve scope, event time, maintenance state and uptime; inspect no traffic or device content.

**Authoritative supporting data**

Authoritative sources: current network topology and room-to-access-point mapping, service incidents, maintenance and outage domains; user-reported room and time. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: diagnosis by scope comparison. Check whether the room point, neighbouring points and building service share the event, then classify local, wider, conflicting or not assessable with provenance. Decision interval: the user-specified interval. Use time-aligned mapped sources and eligible controls; retain alternative explanations, report uncertainty, and return no defensible diagnosis if evidence cannot distinguish them.

**Answer boundary**

The service cannot diagnose personal devices or remote services without evidence. If room/asset mapping, service scope, freshness or permission is missing or conflicting, state confirmed facts only and require a manual check or authorised fallback; never promise device-specific performance.

**Why this is useful:** Supports a choice between moving, checking a personal device or reporting a broader authorised service incident.

### AO-050

CORE | L2 - evidence transparency | R2 - integration-dependent

**Academic office occupant question:** Why is this room being recommended for hybrid work, which evidence is current, and which criteria remain unknown?

**Sensors and telemetry**

Room environmental, numerical acoustic and anonymous occupancy evidence used in the recommendation, plus authorised network, power and AV status. Require target-room coverage, mapped service assets, synchronised timestamps, aggregation, calibration, uptime and gap/drift flags; corridor data is context only and no content or audio is inspected.

**Authoritative supporting data**

Authoritative sources: current room access, privacy classification, accessibility, booking, equipment inventory, service incidents and maintenance; the user states the hybrid-work requirements. Owner-controlled room and service records take precedence; show source owner, effective/retrieval times and permission scope.

**Analysis required**

Operation: provenance summary and answerability assessment. List each criterion, source, observation or retrieval time, quality, result and uncertainty; distinguish verified facts, estimates and omitted criteria. State which missing evidence would make a criterion or the recommendation not assessable.

**Answer boundary**

Do not imply that unmeasured criteria are satisfactory or expose infrastructure details beyond permission. If eligibility, mapping, coverage, quality, freshness or permission is missing or conflicting, return confirmed facts only or mark the criterion not assessable.

**Why this is useful:** This allows the occupant to judge a hybrid-work recommendation instead of accepting an unexplained score.

## 6. Transitions, visitors and multi-location commitments

Questions AO-051-AO-052

### AO-051

CORE | L2 - route + buffer calculation | R2 - integration-dependent

**Academic office occupant question:** I have an office meeting, a lecture and another meeting on different floors. Which verified route and transition buffer should I allow?

#### Sensors and telemetry

Authorised lift, door and closure status; proposed aggregate corridor flow where privacy and coverage permit. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: minimum user-authorised locations and times, current room assignments, staff access, accessible route graph, lift availability and temporary closures. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authorised lookup, route calculation and time-buffer recommendation. Compute valid routes, include setup and transition margins, account for current restrictions and show a fallback route. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

Do not inspect unrelated calendar details or guarantee travel time. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Helps the occupant avoid lateness and unrealistic transitions across a multi-floor day.

### AO-052

CORE | L2 - gap-work recommendation | R2 - integration-dependent

**Academic office occupant question:** I have a 45-minute gap before my next commitment. Which approved nearby workspace leaves enough useful time after walking and setup?

#### Sensors and telemetry

room-level environment, anonymous occupancy, numerical acoustics and authorised network/power status in eligible workspaces; lift/route status. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown; service telemetry mapped to the room/access point/circuit/asset; preserve scope, event time, maintenance state and uptime; inspect no traffic or device content.

#### Authoritative supporting data

Authoritative sources: user-authorised next location and time; current workspace permission, availability, opening, accessibility and closures. No task-content inspection. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authoritative lookup, route calculation and recommendation. Reserve travel and setup time, filter to authorised spaces available for the usable remainder, then rank by service and environmental fit. Decision interval: the stated 45-minute gap. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

Do not book a space, infer personal productivity or recommend a room that cannot be exited in time. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Turns a short timetable gap into a realistic work opportunity without risking the next commitment.

## 6. Transitions, visitors and multi-location commitments

Questions AO-053-AO-054

### AO-053

CORE | L2 - accessible visitor route | R2 - integration-dependent

**Academic office occupant question:** A colleague is visiting from another building. What is the clearest currently verified step-free route to our confirmed meeting location?

#### Sensors and telemetry

Authorised entrance, lift and door status; optional aggregate congestion indicators. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current approved meeting location, visitor entrance and access process, step-free route graph, lift status, temporary closures and public/staff boundary information. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authoritative lookup and route calculation. Generate the simplest valid accessible route, describe landmarks, estimate time and provide a verified fallback if a lift or entrance is unavailable. Decision interval: the current state at retrieval time. Expose inputs, units, assumptions and a range where uncertainty affects the decision.

#### Answer boundary

The service cannot grant visitor access or certify accessibility from sensors. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Reduces visitor confusion and supports a clear, inclusive arrival route.

### AO-054

CORE | L2 - disruption route recommendation | R2 - integration-dependent

**Academic office occupant question:** The lift is officially unavailable. What verified step-free alternative, if any, can I use, and how much extra time should I allow?

#### Sensors and telemetry

Authoritative telemetry: lift state and closure events; door and route availability; optional aggregate corridor flow. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current accessible-route graph, staff access, staircase and door status, temporary closures and official accessibility guidance. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authoritative lookup, route calculation and recommendation. Validate the lift report, remove unavailable paths, calculate permitted alternatives and time difference, and identify when no suitable alternative exists. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

Do not direct a user to stairs if they require step-free access, or treat telemetry as accessibility certification. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Supports a practical response to a verified lift outage without improvising an inaccessible route.

## 6. Transitions, visitors and multi-location commitments

Questions AO-055-AO-056

### AO-055

CORE | L2 - visitor-location recommendation | R2 - integration-dependent

**Academic office occupant question:** Where can I meet a visitor without directing them into staff-only or restricted areas?

#### Sensors and telemetry

public-area environment, aggregate footfall and numerical acoustics may support comfort; authorised entrance and lift status. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current visitor policy, approved meeting points, access boundaries, opening hours, accessible routes and closures. Meeting details are supplied minimally by the user. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authoritative lookup and recommendation. Filter to approved visitor locations, compare route simplicity and current conditions, and provide check-in wording and a fallback. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

Do not expose staff-only routes, grant access or reveal sensitive occupancy. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Helps the occupant host a visitor smoothly while respecting staff-only and restricted areas.

### AO-056

CORE | L1 - entrance/access lookup | R2 - integration-dependent

**Academic office occupant question:** Which entrance may I use for an early appointment, and what confirmed fallback applies if its status changes?

#### Sensors and telemetry

Authoritative telemetry: entrance, door and lift status; no additional environmental sensing is required. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current staff access rights, opening hours, approved early-entry process, accessible route, temporary closures and official security guidance. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authoritative lookup and route calculation. Identify entrances actually open to the user at the requested time, calculate the permitted route and show any check-in requirement. Decision interval: the user-specified interval. Expose inputs, units, assumptions and a range where uncertainty affects the decision.

#### Answer boundary

Do not infer access from a door opening event, disclose security-sensitive detail or grant early entry. Conflicting status must be deferred to Security or the authoritative access service. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** This prevents an unnecessary trip to a closed or unauthorised entrance before an early appointment.

## 6. Transitions, visitors and multi-location commitments

Questions AO-057-AO-058

### AO-057

USEFUL | L2 - congestion forecast | R2 - integration-dependent

**Academic office occupant question:** At 10:55, is class-change foot traffic likely to make my route materially slower than usual?

#### Sensors and telemetry

privacy-preserving corridor and stairway counts at relevant route segments, lift state and door events, aggregated without individual tracking. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current teaching-transition timetable at aggregate level, route graph, temporary closures and access constraints. No student identities or class attendance. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: short-horizon flow forecast and route calculation. Compare current and historical transition peaks for the route, estimate additional time and suggest an authorised alternative if materially better. Decision interval: the route window around 10:55. Use eligible comparable history and known schedules; report prediction range, confidence and recheck time, or fall back to confirmed status.

#### Answer boundary

Do not guarantee travel time or expose named class movements. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Helps the occupant leave at a realistic time when class-change congestion may affect the route.

### AO-058

USEFUL | L2 - equipment-move route | R2 - integration-dependent

**Academic office occupant question:** I need to move authorised equipment between rooms. Which valid step-free route avoids confirmed closures without assuming I may move the equipment?

#### Sensors and telemetry

Authoritative telemetry: lift, door, corridor and closure status; optional broad congestion indicators. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current permitted origin/destination access, step-free route graph, lift dimensions/status and temporary works. Equipment-movement permission and handling rules remain separate authoritative inputs. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: route calculation with explicit preconditions. Compute a feasible step-free path and timing only after the user confirms that movement is authorised; provide fallback or not assessable if dimensions/status are unknown. Decision interval: the user-specified interval. Expose inputs, units, assumptions and a range where uncertainty affects the decision.

#### Answer boundary

The service cannot approve equipment movement, lifting practice or safety. It must not expose security-sensitive routes. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Supports route planning while keeping equipment permission, handling and safety with the responsible owner.

## 6. Transitions, visitors and multi-location commitments

Questions AO-059-AO-060

### AO-059

USEFUL | L2 - privacy-minimised day plan | R2 - integration-dependent

**Academic office occupant question:** Can you sequence several building commitments using only the times and locations I choose to share?

#### Sensors and telemetry

workspace environment and route-flow evidence only for candidate spaces; authorised lift/closure status. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: only user-authorised start/end times, locations, durations and hard access needs; current room/access status, opening, closures and route graph. Sensitive titles can remain hidden. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: schedule synthesis and route recommendation. Use minimal event metadata, calculate buffers, insert optional workspace periods and show the exact data fields used. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

Do not retrieve or expose event titles, participants or notes unless strictly necessary and explicitly authorised. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Provides a building-aware day plan without broad calendar surveillance.

### AO-060

CORE | L2 - room-change verification | R2 - integration-dependent

**Academic office occupant question:** I received a room-change message. Can you verify the new location in the owning system and show the route, access constraints and setup time?

#### Sensors and telemetry

Authoritative telemetry: room-assignment change event, access, lift and closure status; proposed room service/environment evidence for the verified new location. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current authorised timetable or booking system, official notification source, room purpose, staff access, accessibility, equipment/service inventory and route graph. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: source verification, authorised lookup and route calculation. Confirm the change against the owning system, then compute route and setup implications and identify unresolved service checks. Decision interval: the user-specified interval. Expose inputs, units, assumptions and a range where uncertainty affects the decision.

#### Answer boundary

Do not act on an unauthenticated message, disclose another booking or grant access. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** This protects the occupant from acting on an unverified room change while preserving enough time to adapt.

# 7. Accessibility, inclusion, well-being and irregular working

Questions AO-061-AO-062

## AO-061

CORE | L2 - personalised constraints | R2 - integration-dependent

**Academic office occupant question:** Which authorised workspace best meets my stated step-free, low-noise and low-crowding needs today?

### Sensors and telemetry

room-level numerical acoustic indicators and anonymous aggregate occupancy in eligible workspaces; authorised lift, door and closure status. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

### Authoritative supporting data

Authoritative sources: current workspace permission and availability, accessible-route and room-feature data, lift status, opening hours and user-stated constraints. Accessibility records come from competent sources. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

### Analysis required

Operation: authoritative lookup, observation and multi-criteria recommendation. Apply step-free access and permission as hard constraints, then rank verified spaces by current or forecast noise and crowding with uncertainty. Decision interval: the requested period today. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

### Answer boundary

The service cannot certify accessibility from sensors or infer a disability. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Supports an inclusive workspace choice without requiring unnecessary personal disclosure.

## AO-062

CORE | L2 - dynamic accessible route | R2 - integration-dependent

**Academic office occupant question:** Using current lift, door and closure status, what is the best verified step-free route that avoids unusually busy corridors?

### Sensors and telemetry

Authoritative telemetry: lift, door and closure status; proposed aggregate corridor counts where coverage is valid. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

### Authoritative supporting data

Authoritative sources: current step-free route graph, access permissions, temporary works and user-specified origin/destination. Accessibility authority data takes precedence. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

### Analysis required

Operation: authoritative lookup, route calculation and congestion-aware recommendation. Remove invalid paths, compare remaining routes by time and aggregate crowding, and provide a verified fallback. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

### Answer boundary

Do not suggest stairs as a fallback to a step-free requirement, or guarantee that a route will remain uncrowded. Missing lift/door status requires a cautious route or referral. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** This makes a step-free journey more predictable using current authoritative route status.

## 7. Accessibility, inclusion, well-being and irregular working

Questions AO-063-AO-064

### AO-063

USEFUL | L1 - approved break-space lookup | R2 - integration-dependent

**Academic office occupant question:** Is there a confirmed quiet break space near my office that I am permitted to use now?

#### Sensors and telemetry

numerical acoustic and broad occupancy indicators in approved break areas may provide current context. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: break-space classification, eligibility, opening, access, accessibility, privacy and closure data; current route and lift status. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authoritative lookup and route recommendation. Identify approved options, show opening/access and route, and use environmental evidence only where appropriate and non-sensitive. Decision interval: the current state at retrieval time. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

Do not infer that an apparently empty room is a rest space, expose occupancy of sensitive facilities or provide medical advice. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Helps the occupant take a restorative break without entering an unsuitable or restricted space.

### AO-064

USEFUL | L2 - light-stability recommendation | R3 - research-grade or restricted

**Academic office occupant question:** I am sensitive to glare and strong lighting. Which authorised workspace is likely to have the most stable measured light this afternoon?

#### Sensors and telemetry

desk-plane and vertical illuminance at representative positions, daylight/solar reference and optional authorised blind state. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only.

#### Authoritative supporting data

Authoritative sources: current eligible workspace and desk-zone permissions, façade/shading geometry, accessibility needs, room opening and weather forecast. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: observation, forecast and comparison. Compare only validly instrumented permitted zones, estimate illuminance variability and glare risk, and rank options with confidence and a recheck time. Decision interval: the requested period this afternoon. Use eligible comparable history and known schedules; report range, confidence and recheck time, apply permission/accessibility/privacy/safety constraints before ranking, and provide only a verified fallback.

#### Answer boundary

Do not infer a health condition, use cameras or recommend an unallocated desk. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports a better visual environment for sustained work while keeping personal disclosure minimal.

## 7. Accessibility, inclusion, well-being and irregular working

Questions AO-065-AO-066

### AO-065

CORE | L1 - early-hours readiness lookup | R2 - integration-dependent

**Academic office occupant question:** I need to start early tomorrow. Which entrances, lifts and office areas are officially confirmed as available to me?

#### Sensors and telemetry

Authoritative telemetry: entrance, door, lift, access-control and closure status; optional room environmental state only after access is confirmed. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current staff permissions, opening hours, approved early-working process, office allocation, lone-working rules, planned works and official security contacts. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authorised lookup. Verify each access component for the requested time, list confirmed route and workspace status, and identify any mandatory check-in or restriction. Decision interval: the stated period tomorrow. State source owner, effective/version time, retrieval time and conflicts; do not extrapolate beyond the record.

#### Answer boundary

Do not infer access from recent door use, grant permission or replace lone-working/security policy. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** This allows the occupant to plan early work without arriving at a closed or unauthorised entrance.

### AO-066

CORE | L1 - late/lone-working rule lookup | R2 - integration-dependent

**Academic office occupant question:** I expect to work late. Which current access, lone-working and check-in requirements apply to the location I plan to use?

#### Sensors and telemetry

No behavioural monitoring is required. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current staff permissions, local lone-working policy, approved check-in process, location-specific restrictions, planned closures and emergency contacts. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authoritative lookup and compliance-oriented checklist. Resolve the exact location and period, present the current requirements and confirm only whether official systems are available. Decision interval: the user-specified interval. State source owner, effective/version time, retrieval time and conflicts; do not extrapolate beyond the record.

#### Answer boundary

The service cannot approve lone working, monitor the user for safety or replace formal check-in and emergency procedures. Missing or conflicting policy requires direct referral to the responsible authority. Official instructions and closures prevail. If authority, location, timing or permission is missing, stale or conflicting, make no clearance claim and defer to the responsible service.

**Why this is useful:** Helps the occupant understand legitimate late-working conditions before committing to the plan.

## 7. Accessibility, inclusion, well-being and irregular working

Questions AO-067-AO-068

### AO-067

USEFUL | L2 - long-session condition warning | R2 - integration-dependent

**Academic office occupant question:** Can you warn me when measured office conditions are becoming unsuitable for my planned long session without judging my health or productivity?

#### Sensors and telemetry

in-room temperature, RH, NDIR CO2, PM2.5, illuminance and numerical acoustics, with one-minute aggregation, calibration and health flags. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: numerical sound only, with no speech or raw audio.

#### Authoritative supporting data

Authoritative sources: current room and valid sensor mapping, user-selected session duration and non-medical ranges, building incidents and official safety messages. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: observation and rule-based warning. Track duration and trend against the declared ranges, distinguish transient from sustained conditions and suggest an approved break, recheck or alternative. Decision interval: the user-specified interval. Report the observation window, aggregation, units, freshness, valid coverage and excluded channels.

#### Answer boundary

Do not infer health, fatigue, stress or productivity, and do not treat operational ranges as medical or safety thresholds. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports responsible pacing during a long desk session without medical or productivity judgement.

### AO-068

CORE | L2 - resilience recommendation | R2 - integration-dependent

**Academic office occupant question:** If my usual room or lift becomes unavailable, which verified route and authorised workspace fallback still meets my stated needs?

#### Sensors and telemetry

Authoritative telemetry: lift, door, closure and room-status events; proposed environment, occupancy and service evidence for eligible alternatives. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current accessible route graph, workspace permission and availability, staff access, opening, accessibility features, maintenance and official incident status. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: contingency lookup, route calculation and recommendation. Remove unavailable elements, identify valid alternatives and present travel, access and service trade-offs with source timestamps. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

Do not assume a substitute route meets the users needs or that an empty room is authorised. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Provides a practical fallback when a room or vertical route becomes unavailable.

## 7. Accessibility, inclusion, well-being and irregular working

Questions AO-069-AO-070

### AO-069

USEFUL | L2 - recurring barrier analysis | R2 - integration-dependent

**Academic office occupant question:** When is my office area usually most crowded or noisy, and which accessible authorised alternative is best supported?

**Sensors and telemetry**

aggregate occupancy, corridor counts and numerical acoustic indicators in the office area and candidate alternatives; no speech or identity data. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

**Authoritative supporting data**

Authoritative sources: user-supplied relevant time windows, current accessible-route and workspace-permission data, room opening, capacity, events and closures. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: historical comparison, forecast and recommendation. Identify recurring peaks, show variability and confidence, then rank only authorised accessible alternatives for those periods. Decision interval: the user-specified interval. Use eligible comparable history and known schedules; report range, confidence and recheck time, apply permission/accessibility/privacy/safety constraints before ranking, and provide only a verified fallback.

**Answer boundary**

Do not infer individual schedules or guarantee a quiet alternative. If destination, access, accessible-route or lift/closure status is missing, stale, conflicting or unauthorised, provide confirmed route facts only, ask for clarification or mark the recommendation not assessable.

**Why this is useful:** Helps the occupant avoid predictable crowding or noise while preserving accessibility and privacy.

### AO-070

CORE | L2 - answerability explanation | R1 - practical

**Academic office occupant question:** When a well-being question cannot be answered reliably, can you explain the evidence gap and direct me to appropriate support instead of giving medical advice?

**Sensors and telemetry**

sensor inventory, location, calibration, uptime, quality and observed-property metadata; no new personal or health sensing.

**Authoritative supporting data**

Authoritative sources: current permissions, approved well-being information boundaries and referral routes; the users question and minimal location/time context. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: answerability assessment. Map the question to required evidence, identify missing or invalid elements, return the confirmed environmental observations and explain why a health or well-being conclusion is not supported. Decision interval: the user-specified interval. State evidence provenance, assumptions, uncertainty and the condition that would make the result not assessable.

**Answer boundary**

Do not infer diagnosis, mental state, disability, fatigue or productivity. The response must distinguish environmental data from health advice and direct urgent or personal concerns to appropriate services. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** This builds trust by exposing evidence limits and directing the occupant to suitable human support.

## 8. Disruptions, reliability, safety and sustainable office use

Questions AO-071-AO-072

### AO-071

CORE | L2 - official-status + fallback | R2 - integration-dependent

**Academic office occupant question:** There is a reported leak or maintenance issue near my office. Is the area officially open, and which authorised alternative workspace is available?

#### Sensors and telemetry

Authoritative telemetry: leak, maintenance, closure and safety status; optional water/leak sensors as secondary evidence; proposed environment/service evidence for alternatives. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; timestamped authoritative route, lift, door and closure status, with uncovered or stale segments shown.

#### Authoritative supporting data

Authoritative sources: current incident record, area closure, access restrictions, staff workspace eligibility, bookability, accessibility, route and official communications. Current Facilities, Security or Safety records and official instructions valid for the interval take precedence over telemetry and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: authoritative lookup and recommendation. Verify area status, do not infer permission from sensor normality, then identify eligible alternatives and a safe route with source timestamps. Decision interval: the user-specified interval. Apply permission, access, accessibility, privacy and safety before ranking; disclose criteria, unknowns and only a verified fallback.

#### Answer boundary

Environmental or leak telemetry cannot override an official closure or certify re-entry. The service cannot inspect or clear the incident. Official instructions and closures prevail. If authority, location, timing or permission is missing, stale or conflicting, make no clearance claim and defer to the responsible service.

**Why this is useful:** Helps the occupant continue work without entering an affected area or relying on informal reports.

### AO-072

CORE | L3 - local environmental diagnosis | R3 - research-grade or restricted

**Academic office occupant question:** My office is unusually hot. Does the evidence support a persistent local service problem, a brief environmental change, or no defensible diagnosis?

#### Sensors and telemetry

calibrated room temperature, optional globe temperature, anonymous occupancy, outdoor weather, supply temperature/airflow, valve or damper state mapped to the served zone, and neighbouring room data only where separately valid. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history.

#### Authoritative supporting data

Authoritative sources: current HVAC asset/served-zone mapping, setpoint and operating schedule where authorised, maintenance state, room use and reported interval. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: observation, matched comparison and cautious diagnosis. Verify persistence, compare with weather and mapped peer rooms, align BMS states and rank local-service, shared-zone or transient explanations. Decision interval: the user-specified interval. Use time-aligned mapped sources and eligible controls; retain alternative explanations, report uncertainty, and return no defensible diagnosis if evidence cannot distinguish them.

#### Answer boundary

Do not operate HVAC, assign blame or claim causality from correlation. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports a choice to wait, relocate temporarily or submit a technically useful facilities report.

## 8. Disruptions, reliability, safety and sustainable office use

Questions AO-073-AO-074

### AO-073

USEFUL | L2 - service-reliability summary | R2 - integration-dependent

**Academic office occupant question:** Which services used by my office had verified reliability problems this month, and should that affect where I work?

**Sensors and telemetry**

Authorised incident and uptime records for network, power, lifts, AV/shared equipment and HVAC; proposed room environment and occupancy for workspace context. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; service telemetry mapped to the room/access point/circuit/asset; preserve scope, event time, maintenance state and uptime; inspect no traffic or device content.

**Authoritative supporting data**

Authoritative sources: current service-area and room mapping, official incident/maintenance records, valid open hours, staff workspace permissions and user-selected critical services. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: historical calculation, comparison and recommendation. Summarise verified incident frequency and duration by service area, account for planned works, and rank alternative workspaces by the users critical requirements. Decision interval: the stated one-month historical window. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

**Answer boundary**

Do not infer service failure solely from user complaints or expose security-sensitive infrastructure. Past reliability does not guarantee future availability. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant select a workspace that is resilient for the services their tasks require.

### AO-074

CORE | L1 - authoritative emergency message check | R2 - integration-dependent

**Academic office occupant question:** Can you verify whether this alarm, closure or evacuation message is authoritative and show the current official instruction?

**Sensors and telemetry**

Authoritative safety telemetry: alarm and authorised emergency/closure communications only; access, lift or door state may support route information after the official instruction.

**Authoritative supporting data**

Authoritative sources: current official emergency channels, Security/Safety instructions, building and area identifiers, time-stamped message source and approved emergency contacts. Current Facilities, Security or Safety records and official instructions valid for the interval take precedence over telemetry and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: source verification and authoritative lookup. Confirm whether the message matches an official channel, present the latest instruction verbatim or faithfully summarised, and identify the owning authority. Decision interval: the user-specified interval. State source owner, effective/version time, retrieval time and conflicts; do not extrapolate beyond the record.

**Answer boundary**

The conversational service must not delay evacuation, reinterpret an alarm, declare an area safe or substitute for emergency services. Official instructions and closures prevail. If authority, location, timing or permission is missing, stale or conflicting, make no clearance claim and defer to the responsible service.

**Why this is useful:** Reduces confusion while preserving the primacy of certified alarms and competent emergency instructions.

## 8. Disruptions, reliability, safety and sustainable office use

Questions AO-075-AO-076

### AO-075

USEFUL | L2 - avoidable-use observation | R2 - integration-dependent

**Academic office occupant question:** Is heating, cooling or lighting operating while this office is estimated to be empty, and what approved non-control action can I take?

**Sensors and telemetry**

Anonymous room occupancy, lighting state or power, HVAC airflow/valve/fan status, room temperature and any mapped room or served-zone energy point. Require synchronised event or one-minute records, verified mapping, calibration/commissioning, uptime and gap/drift flags; whole-building totals cannot support a room-level claim.

**Authoritative supporting data**

Authoritative sources: current room allocation and operating schedule, verified BMS/lighting mapping, permitted occupant actions and official energy guidance. Owner-controlled records valid for the interval take precedence; show source owner, effective/retrieval times and permission scope, without using individual presence history.

**Analysis required**

Operation: calculation and recommendation. Identify sustained periods when anonymous occupancy is estimated as zero while services remain active, quantify the evidence, and suggest only approved non-control actions such as reporting. State the interval, uncertainty and unknowns; never treat the estimate as proof of vacancy.

**Answer boundary**

Occupancy estimates may miss people, and some services must run in empty rooms. Do not switch systems, override controls or identify occupants. If mapping, coverage, quality, time alignment or permission is inadequate, return confirmed facts only or mark the result not assessable.

**Why this is useful:** Gives the occupant a proportionate way to reduce avoidable energy use without taking over building control.

### AO-076

USEFUL | L2 - meeting-space sustainability trade-off | R2 - integration-dependent

**Academic office occupant question:** Would requesting a smaller approved room plausibly reduce this meeting's energy use without weakening access, privacy or comfort?

**Sensors and telemetry**

room-level environment and anonymous occupancy; mapped room or zone HVAC, lighting and plug-load energy where available. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; room-level energy only from a mapped meter/BMS point, not whole-building totals.

**Authoritative supporting data**

Authoritative sources: current authorised room availability, capacity, access, accessibility, equipment, meeting requirements, HVAC zones and energy provenance. Current owner-controlled room, access, IT, power and equipment records valid for the interval take precedence over environmental inference and apparent activity. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: calculation, comparison and recommendation. Check meeting fit and access first, estimate marginal energy only where valid, compare environmental forecasts and present the trade-off and uncertainty. Decision interval: the user-specified interval. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

**Answer boundary**

Do not book the room, treat smaller as automatically more efficient or compromise accessibility/privacy. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports a lower-impact meeting choice without sacrificing access, privacy or environmental suitability.

## 8. Disruptions, reliability, safety and sustainable office use

Questions AO-077-AO-078

### AO-077

USEFUL | L2 - daylight availability forecast | R3 - research-grade or restricted

**Academic office occupant question:** When is daylight likely to be sufficient for desk work here without assuming the building should change its lighting controls?

#### Sensors and telemetry

desk-plane illuminance at the occupied position, outdoor solar radiation and authorised lighting-state context; correct placement, calibration and one-minute aggregation. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: room-level energy only from a mapped meter/BMS point, not whole-building totals.

#### Authoritative supporting data

Authoritative sources: current desk/workspace permission, façade and shading geometry, weather forecast, opening hours and approved occupant lighting guidance. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: observation and daylight forecast. Estimate intervals when daylight contribution meets the declared task-light criterion, show variability and recommend a check rather than automatic control. Decision interval: the user-specified interval. Use eligible comparable history and known schedules; report prediction range, confidence and recheck time, or fall back to confirmed status.

#### Answer boundary

Do not switch lighting, infer visual ability or claim glare-free conditions from horizontal lux alone. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Supports a modest occupant-controlled daylight choice without unsupported control changes.

### AO-078

USEFUL | L3 - intervention evaluation | R3 - research-grade or restricted

**Academic office occupant question:** After the recent maintenance intervention, have office conditions measurably improved once weather, occupancy and operating changes are considered?

#### Sensors and telemetry

room temperature, CO2, RH, relevant airflow/BMS points, anonymous occupancy and outdoor weather, with stable pre/post calibration, timestamps and data-quality metadata. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; room-level energy only from a mapped meter/BMS point, not whole-building totals.

#### Authoritative supporting data

Authoritative sources: intervention description and time, served-zone/asset mapping, monitoring-validity periods, room use, closures and comparable control rooms if legitimately available. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

#### Analysis required

Operation: pre/post comparison or interrupted time-series evaluation. Use comparable operating periods, adjust for weather and occupancy, quantify effect and uncertainty, and test whether improvement persisted. Decision interval: defined pre- and post-intervention windows. Align clocks and windows, use eligible comparators and relevant controls, and report effect and uncertainty without claiming causality.

#### Answer boundary

Do not claim causality if other changes coincide, expose maintenance-sensitive data beyond permission or judge contractor performance. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Gives the occupant a defensible view of whether the reported office condition has improved.

## 8. Disruptions, reliability, safety and sustainable office use

Questions AO-079-AO-080

### AO-079

CORE | L2 - observability audit | R1 - practical

**Academic office occupant question:** Which missing, stale or unreliable measurements currently prevent dependable answers about office comfort?

**Sensors and telemetry**

sensor inventory, observed property, room/served-zone location, sampling, calibration, uptime, gaps, drift and maintenance metadata; no new personal sensing.

**Authoritative supporting data**

Authoritative sources: current room purpose and access, HVAC and spatial mapping, supported question types and minimum-evidence rules. Proposed sensors remain clearly labelled as absent until verified. Current owner-controlled room purpose, allocation/booking, access, accessibility, privacy, opening, closure and service records valid for the interval take precedence over sensor inference and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: answerability and coverage assessment. Map common office questions to required variables and locations, classify each as supported, partially supported or not assessable, and list the missing evidence without inventing readings. Decision interval: the current state at retrieval time. State evidence provenance, assumptions, uncertainty and the condition that would make the result not assessable.

**Answer boundary**

Do not equate dense corridor sensing with room coverage or assume metadata means a device is operational. Security-sensitive asset detail must remain access-controlled. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Shows which office questions are currently supportable and where additional sensing or integration is required.

### AO-080

CORE | L2 - evidence package | R2 - integration-dependent

**Academic office occupant question:** Can you prepare a concise, time-stamped evidence summary of this recurring office problem for a facilities report without diagnosing the cause or assigning blame?

**Sensors and telemetry**

relevant room-level environmental, acoustic, occupancy and mapped BMS/service telemetry, with timestamps, units, calibration, uptime, gaps and source provenance. Quality: commissioned target-room/desk-zone or validated served-zone coverage; synchronised timestamps, stated aggregation, calibration, uptime, gaps/drift and relocation flags. Corridor or adjacent-room readings are context only. Privacy/service guard: anonymous counts with stated uncertainty and no identity or movement history; numerical sound only, with no speech or raw audio.

**Authoritative supporting data**

Authoritative sources: user-defined issue, location and period; current room and asset mapping; authorised incident and maintenance references; official reporting route. No emails, conversations or named occupant histories. Current destination, route graph, access/accessibility, lift/door and closure records valid for the interval take precedence over footfall and the 2021 plan. Retain source owner, effective and retrieval times, permission scope and conflicts.

**Analysis required**

Operation: calculation and evidence summarisation. Describe observed pattern, recurrence, comparable context, data quality and unresolved questions; attach concise tables or plots if allowed and distinguish observation from possible explanation. Decision interval: the user-specified interval. Expose inputs, units, assumptions and a range where uncertainty affects the decision.

**Answer boundary**

Do not assert a fault, cause, compliance breach or responsible party. Suppress sensitive operational details and state when room coverage or data quality makes the issue not assessable. If eligibility, permission, room or served-zone coverage, calibration/health, time alignment or comparator evidence is missing, stale or conflicting, ask for clarification, return confirmed facts only or mark the result not assessable; never substitute corridor or adjacent-room data.

**Why this is useful:** Helps the occupant submit a clear, proportionate evidence summary that facilities staff can investigate.

# 4. Catalogue summary and implementation requirements

The catalogue combines everyday office-work questions with room, allocation, access, service and route integrations, plus carefully bounded comparison, forecast, diagnosis and safety-sensitive cases. Readiness identifies the sensing, authority, quality, permission and safe-failure conditions required before a responsible answer can be given.

| | | |
|---|---|---|
| **Questions** | **80** | Eight academic-office-occupant-centred sections containing ten questions each. |
| **R1 practical** | **2** | Can be supported with proposed core sensing and basic current spatial, service or provenance feeds. |
| **R2 integration-dependent** | **62** | Require room, allocation, booking, access, accessibility, IT, BMS, maintenance, visitor, lone-working or forecasting integration. |
| **R3 research or restricted** | **16** | Require dense coverage, validated comparison/diagnosis, controlled operational data, or stronger privacy and safety controls. |

## Recurring system capabilities

**1 Workday and focus planning**

Resolve focused-work blocks, meetings, supervision, hybrid calls, visitors, transitions, breaks, contingencies and irregular-hour constraints.

**2 Room, allocation and permission reasoning**

Use current room purpose, workspace allocation, bookability, staff access, accessibility, privacy classification, routes, lifts, opening and closures.

**3 Environmental and service evidence**

Combine correctly located room or desk-zone sensing with anonymous occupancy, BMS, power, network, shared-resource and authorised service status.

**4 Forecast, comparison and diagnosis**

Predict conditions and flow, build matched baselines, compare eligible spaces, reconstruct service events and avoid unsupported causal claims.

**5 Evidence quality and provenance**

Check room/asset mapping, sampling, clocks, commissioning/calibration, uptime, gaps, drift, source agreement, processing lineage and permissions.

**6 Privacy, accessibility, safety and abstention**

Minimise personal and work-content data, refuse unauthorised access or safety claims, preserve inclusive alternatives and return not assessable when evidence is inadequate.

### QUESTION RECORD STRUCTURE

Each entry can become a system test containing the natural utterance and paraphrases, required entities, minimum room, workspace and service evidence, source-precedence rule, quality threshold, expected operation type, permission rule, answerability class, acceptable clarification, uncertainty statement, safe failure, expected answer characteristics and prohibited overclaims.

# 5. Evidence, permissions and answerability rules

These rules apply across all 80 records and define when the building may answer, qualify, ask for clarification, defer to an authorised service or return not assessable.

### Building and room status

The May 2021 plan is a historical spatial reference. Current room purpose, ownership, allocation, capacity, staff access, bookability, equipment, accessible features, entrances, lifts, services and closures require an authoritative current source.

### Spatial coverage

A room-level answer requires a correctly located sensor inside that room or a validated served zone. Corridor and adjacent-room data may be labelled context but cannot be silently substituted.

### Time window and operation type

Every answer resolves the requested interval and labels its operation as observation, authorised lookup, calculation, estimate, forecast, comparison, diagnosis or recommendation. Preserve observed-at and retrieved-at times.

### Calibration and data quality

Use stated sampling and aggregation, synchronised clocks, commissioning/calibration, uptime, gap and drift flags, maintenance state and source agreement. Failed, stale, relocated or unverified points cannot support a precise claim.

### Work-content minimisation

Do not inspect emails, documents, calendars beyond user-authorised scheduling context, meetings, calls, research or teaching content, screens, speech, network traffic or device content. Process only the minimum constraints needed.

### Occupancy, sound and privacy

Use anonymous aggregate counts and numerical LAeq or percentile statistics. Do not identify people, retain speech, infer conversation or presence history, track movement or estimate performance or productivity.

### Environmental interpretation

Direct-reading NDIR $CO_2$ is a ventilation indicator, not complete air quality. Temperature, RH, PM, indicative TVOC, airflow, radiant conditions, light, outdoor weather and activity may be relevant subject to coverage and quality.

### Office and shared-space authority

Environmental IoT data cannot authorise room or desk use, certify confidentiality or accessibility, or prove a shared resource is available. Allocation, privacy, booking, induction and equipment status come from authoritative systems.

### Operational services and permissions

Room allocation, bookings, access, visitors, lifts, network, power, shared equipment, maintenance and closures come from authorised feeds. The service cannot book, grant access, operate plant or override responsible teams.

### Emergency and high-consequence answers

Certified alarms and official communications take precedence. Safety, health, lone-working, accessibility and compliance questions require competent authority; missing or conflicting critical evidence requires a limited or not-assessable answer.

### FINAL ANSWERABILITY RULE

Answer only when the required variables, current room or served-zone coverage, requested interval, source authority, quality threshold, permissions, accessibility and safety constraints are adequate. Otherwise ask for clarification, omit unsupported criteria, return confirmed facts with explicit uncertainty, restrict disclosure, defer to the responsible service, or mark the analytical part not assessable.